**Practical Name:** Practical 14: Selenium Web Automation

**Aim:** To create Selenium automation scripts in Python for browser navigation, form handling, web element interaction and validation testing.

Source Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()

wait = WebDriverWait(driver, 10)

try:
    # Browser Navigation
    driver.get("https://www.saucedemo.com/")

    print("Page Title:", driver.title)

    # Form Handling
    wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
        )
    ).send_keys("standard_user")

    driver.find_element(
        By.ID, "password"
    ).send_keys("secret_sauce")

    # Web Element Interaction
    driver.find_element(
        By.ID, "login-button"
```

```
    ).click()

    # Validation
    wait.until(
        EC.url_contains("inventory")
    )

    assert "inventory" in driver.current_url

    print("Login Validation: PASS")

    # Product Interaction
    wait.until(
        EC.element_to_be_clickable(
            (By.ID, "add-to-cart-sauce-labs-backpack")
        )
    ).click()

    print("Product Selection: PASS")

    # Cart Navigation
    driver.find_element(
        By.CLASS_NAME,
        "shopping_cart_link"
    ).click()

    assert "cart" in driver.current_url

    print("Cart Navigation: PASS")
    print("All Selenium Web Automation Tests Passed")

finally:
    driver.quit()
```